Spontaneous symmetry breaking in two-channel asymmetric exclusion processes with narrow entrances

Reproduction of Pronina & Kolomeisky, J. Phys. A 40, 2275 (2007)

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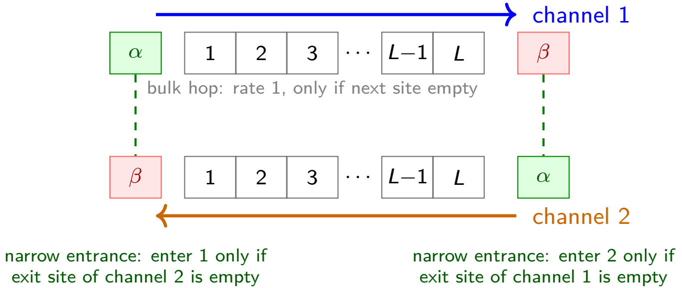
Why study exclusion processes?

- Asymmetric Simple Exclusion Process (TASEP) is the reference model of *non-equilibrium* transport — the role of the Ising model for equilibrium statistical mechanics.
- Minimal description of interacting particles on a 1D lattice with hard-core exclusion.
- Applications: motor proteins on cytoskeletal filaments (kinesin/dynein), ribosome motion in mRNA translation, car traffic, nanopore transport.

Motivation for two channels

In cells, kinesin and dynein walk *in opposite directions* on the same microtubule. Two lanes of particles moving in opposite directions, coupled at the boundaries — can they *break symmetry* spontaneously?

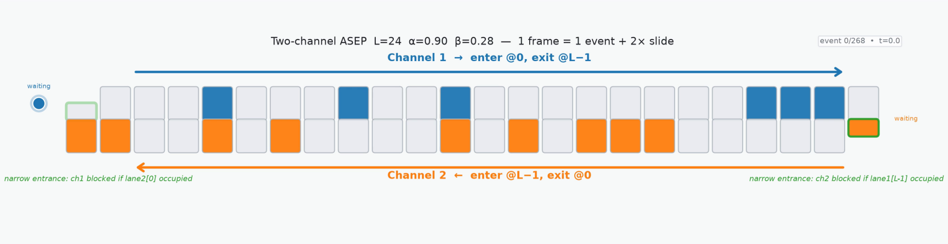
The paper's question



Steady states

Symmetric: $\rho_1 = \rho_2$ (both lanes equal).
Asymmetric: despite exact symmetry of the rules, one lane dense, the other dilute — $\rho_1 \neq \rho_2$.

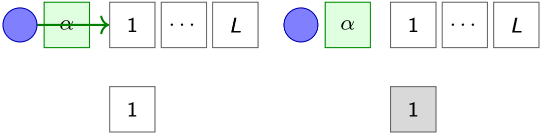
The model: two-channel ASEP with narrow entrances



Click the frame to play the animation (`lattice_animation.mp4`).

The narrow-entrance rule

- Enter channel 1 (rate α) iff site 1 of channel 1 is empty **and** the exit site of channel 2 is empty.
- Enter channel 2 (rate α) iff site L of channel 2 is empty **and** the exit site of channel 1 is empty.



(a) exit of ch.2 empty
$\Rightarrow$ particle enters

(b) exit of ch.2 occupied
$\Rightarrow$ entry blocked

The coupling

Each channel's *entrance* is coupled to the other channel's *exit* — a shared bottleneck at each end.

What we want to compute

Steady-state properties as a function of the rates (α, β):

- $\alpha =$ entrance rate of either channel (identical for both).
- $\beta =$ exit rate of either channel (identical for both).
- Phase diagram in the (α, β) plane.
- Bulk densities ρ_1, ρ_2 and currents J_1, J_2.
- The joint density distribution $P(\rho_1, \rho_2)$ — the diagnostic for spontaneous symmetry breaking (SSB).

Recap: single-lane TASEP

One channel, particles enter with rate α, hop right with rate 1, exit with rate β.

Mean-field approximation

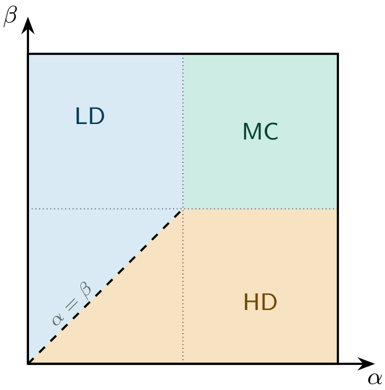
Assume no correlations between sites (eq 1 of the paper):

$$\langle p_i p_j \rangle = \langle p_i \rangle \langle p_j \rangle = p_i p_j.$$

The current across bond $i \to i{+}1$ is then

$$J = \langle p_i (1 - p_{i+1}) \rangle = \rho(1-\rho), \qquad \rho = \langle p_i \rangle.$$

Single-lane phases and phase diagram

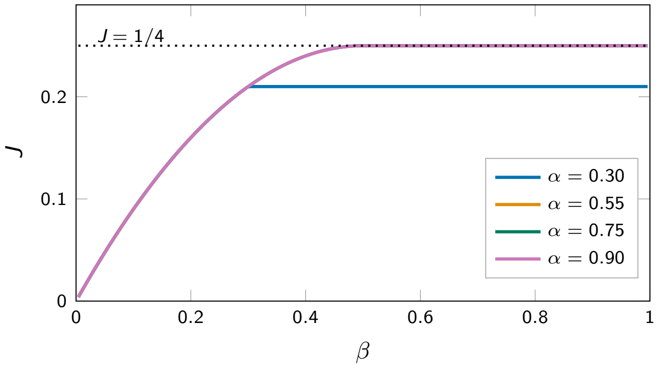


Phase	condition	J, ρ
LD	$\alpha < \beta$, $\alpha < 1/2$	$J{=}\alpha(1{-}\alpha)$, $\rho{=}\alpha$
HD	$\beta < \alpha$, $\beta < 1/2$	$J{=}\beta(1{-}\beta)$, $\rho{=}1{-}\beta$
MC	$\alpha > 1/2$, $\beta > 1/2$	$J{=}1/4$, $\rho{=}1/2$

LD = low density, HD = high density, MC = maximal current.

For a single lane, **mean-field (MFT) is exact** (matrix ansatz).

Single-lane currents



- $J = c(1-c)$ with $c = \min(\alpha, \beta, 1/2)$: rises along $\beta(1-\beta)$, then saturates.
- All curves with $\alpha > 1/2$ collapse onto $J = 1/4$ — the bulk becomes rate-limiting.
- For $\alpha < 1/2$ the curve stays on its own $\alpha(1-\alpha)$ plateau — no overlap.

Strategy: two single-lane TASEPs with effective rates

Two single-lane TASEPs coupled *only through the entrances* (eq 2); the *exit* rate β is identical for both, only the entrance rates differ.

Effective entrance rates

$$\alpha_1 = \alpha \langle 1 - m_1 \rangle = \alpha(1 - m_1),$$
$$\alpha_2 = \alpha \langle 1 - p_L \rangle = \alpha(1 - p_L),$$

$\alpha_1, \alpha_2 =$ entrance rates felt by channels 1, 2; $m_1 =$ channel 2's exit density, $p_L =$ channel 1's exit density.

Each channel is a standard TASEP with rates (α_1, β), (α_2, β) — whose steady states we already know.

Nine candidate phases

Each channel in LD/HD/MC $\Rightarrow 3 \times 3$; filter by self-consistency of α_1, α_2:

ch.2\ch.1	LD	HD	MC
LD	**LD/LD**	**HD/LD**	MC/LD
HD	**LD/HD**	HD/HD	MC/HD
MC	LD/MC	HD/MC	MC/MC

Only the **bold** combinations survive.

Symmetric phases: set-up

Notation: p_i = density at site i of channel 1, m_i = density at site i of channel 2, both counted from their own entrance ($i = 1, \ldots, L$). So m_1 is channel 2's exit density, p_L is channel 1's exit density.

Symmetric ansatz (eq 6-7)

$$J_1 = J_2, \quad \rho_1 = \rho_2, \quad \alpha_1 = \alpha_2, \qquad p_i = m_{L-i+1}.$$

- Both channels in the same single-lane phase.
- Symmetry + the exit current relation determine the effective rates *self-consistently*.
- Key relation (exit current of channel 2, eq 8/11):

$$J_2 = \beta\, m_1.$$

Symmetric MC phase (eqs 8–10)

Step 1: current fixes m_1 (eq 8). In MC, $J_1 = J_2 = 1/4$; the exit current of channel 2 gives

$$J_2 = \beta m_1 = \frac{1}{4} \quad \Rightarrow \quad m_1 = \frac{1}{4\beta}.$$

(Physical range: $\beta > 0$.) **Step 2: effective entrance rate (eq 9).**

$$\alpha_1 = \alpha(1 - m_1) = \alpha\left(1 - \frac{1}{4\beta}\right).$$

Step 3: existence condition (eq 10). MC needs $\alpha_1 > 1/2$ and $\beta > 1/2$:

$$\alpha\left(1 - \frac{1}{4\beta}\right) > \frac{1}{2} \quad \Longrightarrow \quad \alpha > \frac{2\beta}{4\beta - 1}$$

Symmetric LD phase (eqs 11–13)

Step 1: current fixes m_1 (eq 11). In LD, $J_1 = J_2 = \alpha_1(1-\alpha_1)$; exit current of channel 2:

$$J_2 = \beta m_1 = \alpha_1(1-\alpha_1) \quad \Rightarrow \quad m_1 = \frac{\alpha_1(1-\alpha_1)}{\beta}.$$

Step 2: self-consistency for α_1 (eq 2).

$$\alpha_1 = \alpha(1-m_1) = \alpha\left(1 - \frac{\alpha_1(1-\alpha_1)}{\beta}\right).$$

A quadratic in α_1; physical root (eq 12):

$$\alpha_1 = \frac{\alpha+\beta-\sqrt{(\alpha+\beta)^2-4\alpha^2\beta}}{2\alpha}.$$

Step 3: existence condition (eq 13). LD needs $\alpha_1 < 1/2$ and $\alpha_1 < \beta$:

$$\alpha < \frac{2\beta}{4\beta-1}$$

Symmetric HD phase: does not exist

- In HD, $J_1 = J_2 = \beta(1-\beta)$. From $J_2 = \beta m_1$:
$$m_1 = 1 - \beta.$$
- Effective entrance rate:
$$\alpha_1 = \alpha(1 - m_1) = \alpha\beta.$$
- HD requires $\alpha_1 > \beta$, i.e. $\alpha\beta > \beta$, i.e. $\alpha > 1$. But $\alpha \leq 1$ by definition.

Conclusion

Symmetric HD phase is impossible in the two-channel model with narrow entrances. Only MC and LD survive as symmetric phases.

Asymmetric phases: MC/HD and MC/LD don't exist

- **MC/HD** (eq 15): requires simultaneously $\beta > 1/2$ (for MC) and $\beta < 1/2$ (for HD) — contradictory.
- **MC/LD** (eqs 16–19): MC needs $\alpha_1 > 1/2$, LD needs $\alpha_2 < 1/2$. Using $J_1 = \beta p_L = 1/4$ and $J_2 = \beta m_1 = \alpha_2(1 - \alpha_2)$ leads to a condition (eq 19) that cannot hold for $0 < \alpha \leq 1$, $0 < \beta \leq 1$.

Conclusion

An asymmetric phase must pair **one channel in LD** with the other in a different phase: the candidate is HD/LD.

Asymmetric HD/LD phase (eqs 20–23)

Conditions (eq 20). Channel 1 in HD: $\alpha_1 > \beta$, $\beta < 1/2$; Channel 2 in LD: $\alpha_2 < \beta$, $\alpha_2 < 1/2$.
Step 1: currents (eq 21).

$$J_1 = \beta p_L = \beta(1-\beta) \quad \Rightarrow \quad p_L = 1-\beta, \qquad J_2 = \beta m_1 = \alpha_2(1-\alpha_2).$$

Step 2: effective rates (eq 22).

$$\alpha_2 = \alpha(1-p_L) = \alpha\beta, \qquad \alpha_1 = \alpha(1-m_1) = \alpha(1-\alpha+\alpha^2\beta).$$

Step 3: existence (eq 23). HD condition $\alpha_1 > \beta$:

$$\alpha(1-\alpha+\alpha^2\beta) > \beta \quad \Longrightarrow \quad \beta < \frac{\alpha}{1+\alpha+\alpha^2}$$

Asymmetric LD/LD phase: set-up (eqs 24–26)

Both channels in LD, but with *different* densities and currents.

Conditions (eq 24)

$\alpha_1 < \beta$, $\alpha_1 < 1/2$ and $\alpha_2 < \beta$, $\alpha_2 < 1/2$.

Currents and effective rates (eqs 25–26)

$$\begin{aligned} J_1 &= \beta p_L = \alpha_1(1-\alpha_1), & J_2 &= \beta m_1 = \alpha_2(1-\alpha_2), \\ p_L &= \frac{\alpha_1(1-\alpha_1)}{\beta}, & m_1 &= \frac{\alpha_2(1-\alpha_2)}{\beta}, \\ \alpha_1 &= \alpha\left(1-\frac{\alpha_2(1-\alpha_2)}{\beta}\right), & \alpha_2 &= \alpha\left(1-\frac{\alpha_1(1-\alpha_1)}{\beta}\right). \end{aligned}$$

Two coupled quadratics in α_1, α_2. Simplify with symmetric and antisymmetric combinations (eq 27):

$$S = \alpha_1 + \alpha_2, \qquad D = \alpha_1 - \alpha_2 \neq 0.$$

Asymmetric LD/LD phase: solve (eqs 27–31)

Step 1: difference of the two equations (eq 28).

$$D = \frac{\alpha}{\beta}\left(\alpha_1 - \alpha_1^2 - \alpha_2 + \alpha_2^2\right) = \frac{\alpha}{\beta} D(1-S).$$

Since $D \neq 0$ (asymmetric phase), we get

$$S = 1 - \frac{\beta}{\alpha} \qquad \text{(eq 29)}$$

Step 2: sum of the two equations (eq 30).

$$S = 2\alpha - \frac{\alpha}{\beta}\left[S - \frac{S^2 + D^2}{2}\right] \qquad \text{(using } \alpha_1^2 + \alpha_2^2 = (S^2 + D^2)/2\text{)}.$$

Step 3: substitute $S = 1 - \beta/\alpha$ (eq 31).

$$D^2 = 1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2$$

Asymmetric LD/LD phase: effective rates (eq 32)

From S and D we reconstruct the two effective entrance rates:

$$\alpha_1 = \frac{S+D}{2}, \qquad \alpha_2 = \frac{S-D}{2}.$$

$$\alpha_1 = \frac{1}{2}\left[1 - \frac{\beta}{\alpha} + \sqrt{1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2}\right],$$

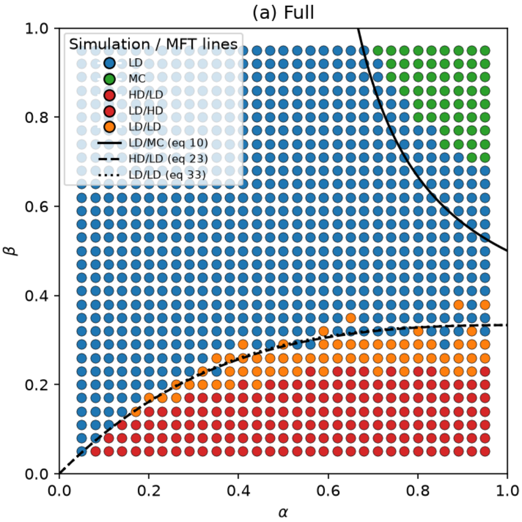
$$\alpha_2 = \frac{1}{2}\left[1 - \frac{\beta}{\alpha} - \sqrt{1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2}\right].$$

Existence (eq 33)

Real roots need $D^2 > 0$; together with the LD conditions (eq 24):

$$\frac{\alpha}{1+\alpha+\alpha^2} < \beta < \frac{\alpha(1-2\alpha) + 2\alpha\sqrt{\alpha^2 - \alpha + 1}}{3}$$

Mean-field phase diagram (summary)



Phase	MFT boundary
MC	$\alpha > \dfrac{2\beta}{4\beta - 1}$ (eq 10)
LD	$\alpha < \dfrac{2\beta}{4\beta - 1}$ (eq 13)
HD	never (needs $\alpha > 1$)
HD/LD	$\beta < \dfrac{\alpha}{1+\alpha+\alpha^2}$ (eq 23)
LD/LD	eq 33 (narrow band)

Transitions: LD$\leftrightarrow$MC and LD$\leftrightarrow$LD/LD are *continuous*; HD/LD$\leftrightarrow$LD/LD is *first order*.

Take stock: what mean-field predicts

1. Two symmetric phases (LD, MC) — HD impossible.
2. Two symmetry-broken phases: HD/LD (first order) and LD/LD (continuous), in a *very narrow* band of β.
3. The phase diagram is fully determined by self-consistency of the effective entrance rates α_1, α_2.

The message to verify

Mean-field predicts *where* transitions happen. Are these positions exact? For single-lane TASEP they are. Here, the boundary coupling may introduce correlations mean-field misses.

High-budget simulation runs

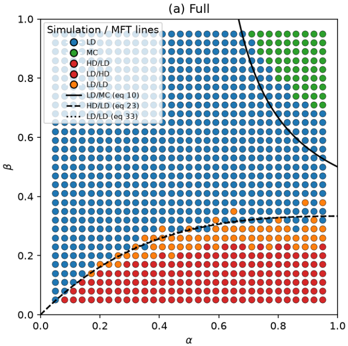
L	steps/point	steps/site
200	1×10^7	5×10^4
500	2.5×10^7	5×10^4
1000	5×10^7	5×10^4
2000	1×10^8	5×10^4

$c =$ prefactor in steps/site

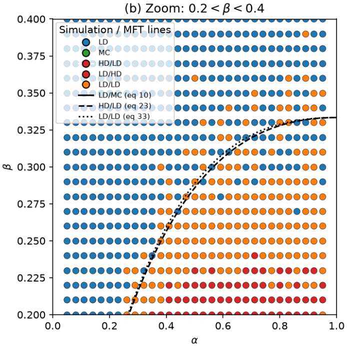
$\approx cL$; here $c \approx 50$ (need $c \approx 100$ for $L \geq 4000$).

- Fenwick-tree BKL, 30 CPU workers (SIM).
- Equilibration budget must **scale with** L (steps/site $\approx c\,L$): a fixed budget under-equilibrates large L.
- Fig 4 scan: 2×10^8 steps/point, 40 β values ($L = 2000$).
- Next: flipping time $\tau \propto e^L$ — exponential growth (Zhu Fig. 3) is the cleanest SSB-is-real fingerprint; needs single long trajectories, CPU-friendly.

Phase diagram: SIM vs MFT ($L = 2000$)



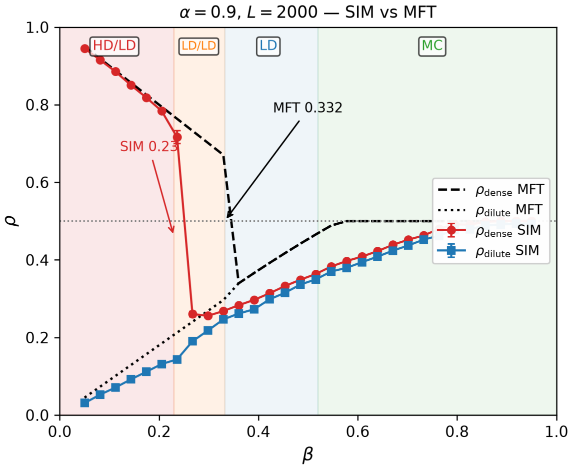
full range — 5×10^4/site



zoom: $0.2 < \beta < 0.4$

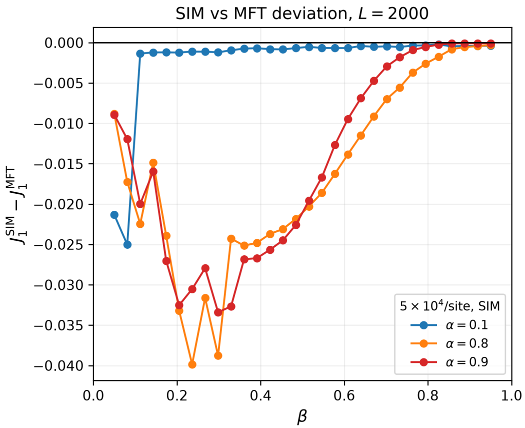
- Lines: MFT (eqs 10, 23, 33); symbols: SIM — topology reproduced (LD, MC, HD/LD, LD/LD).
- **Boundary positions deviate** — the paper's key message (red band below MFT dashed).

Where MFT works — and where it breaks ($\alpha = 0.9$)



- SIM (symbols): sharp HD/LD jump at $\beta \approx 0.23$.
- MFT (dashed): jump at $\beta = 0.332$ (eq 23) — shifted by ~ 0.1.
- Dense plateau $\rho \approx 0.71$ vs MFT ~ 0.8.
- LD at low β and MC at high β agree (see currents backup) — deviation localized at transitions.

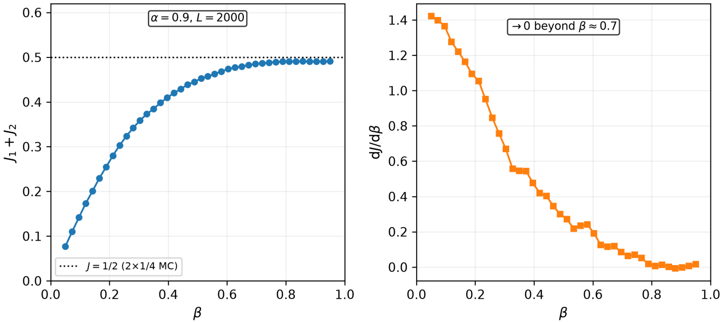
Transition positions: the paper's key message



$J_1^{\mathrm{SIM}} - J_1^{\mathrm{MFT}}$ vs β ($L = 2000$, 5×10^4/site)

- Deviation ≈ 0 deep inside a phase, largest near the transitions (~ 0.04).
- HD/LD boundary: SIM at $\beta \approx 0.23$, MFT at 0.33 (eq 23).
- Unlike single-lane TASEP (MFT exact), here transition positions are only *approximate*.
- Persists at $L = 2000$ — not a finite-size transient.

Total current saturation (paper Fig 4)



- $\alpha = 0.9$, $L = 2000$: $J_1 + J_2$ saturates at the MC value $J = 1/2$ (dotted line), 2×10^8 steps/pt.
- $\mathrm{d}J/\mathrm{d}\beta \to 0$ beyond $\beta \approx 0.7$: the bulk is rate-limiting, entrance coupling irrelevant.

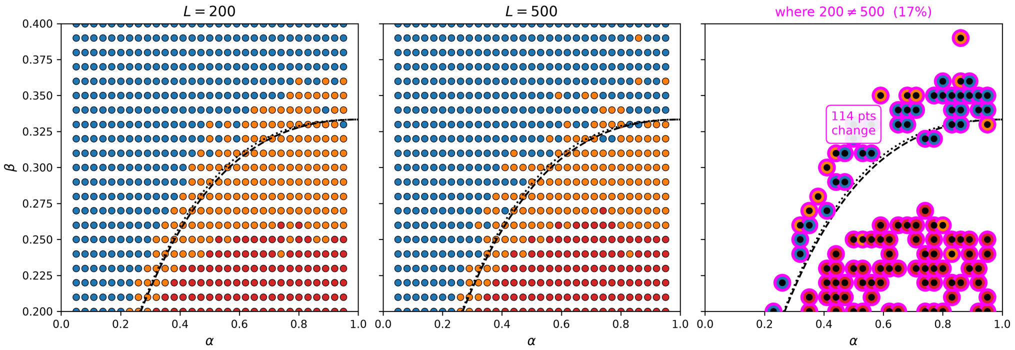
Why does mean-field fail here?

The source of the discrepancy

Mean-field assumes *no correlations* between sites (eq 1). The narrow-entrance coupling ties each entrance to the other channel's exit site — a strongly-correlated, local "impurity" at the boundary that mean-field cannot capture.

- These boundary correlations *propagate into the bulk* and shift the effective phase boundaries.
- The paper's own conclusion (Sec. 4): symmetry breaking in this model is a *boundary-induced* phenomenon.
- The deviation **persists up to** $L = 2000$ at 5×10^4 steps/site — looks like a genuine MFT failure, not a transient; the thermodynamic limit is still open.

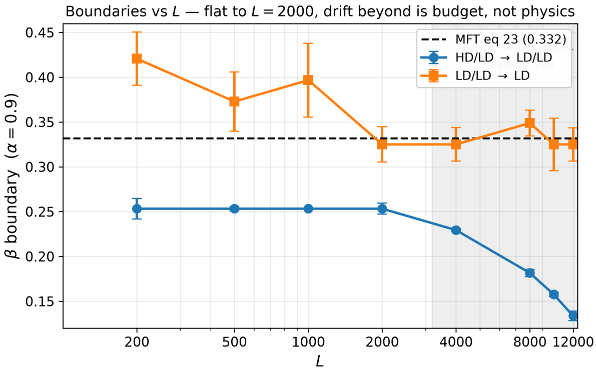
The LD/LD band shrinks with L



Zoom $0.2 < \beta < 0.4$: $L = 200 \to L = 500$ (both 5×10^4/site). Right: magenta = 17% points change phase.

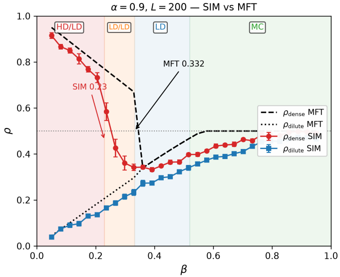
- **Narrowing is discrete** — colour change at same (α, β) = phase change. $L = 500 \to 2000$ is same trend but only $\sim 3\%$, hence subtle (see Fig5).

Phase boundaries vs L (paper Fig 5, $\alpha = 0.9$)

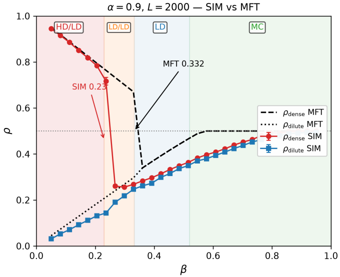


- HD/LD boundary flat at $\beta \approx 0.253$ for $L \leq 2000$ — matches the paper ($\approx$0.26), far from MFT (0.332).
- LD/LD lower boundary scatters around the MFT line.
- Shaded $L \geq 4000$: under-equilibration ($c = 50$; need $c \approx 100 \Rightarrow 8 \times 10^8$ steps at $L = 8000$) — budget caveat.

Finite-size: bulk densities ($\alpha = 0.9$) vs L



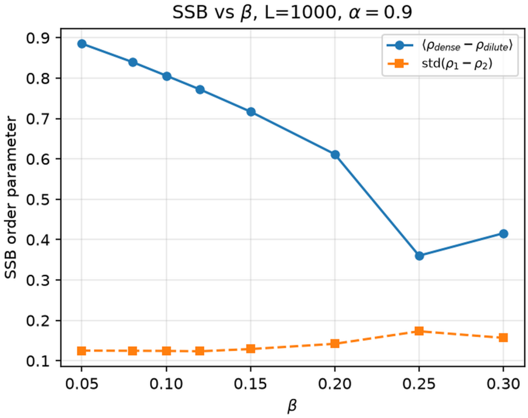
$L = 200$ — broadened



$L = 2000$ — sharp, 5×10^4/site

- HD/LD jump sharpens with L; dilute branch unchanged, error bars collapse.
- Position stable at $\beta \approx 0.23$–0.26 — far from MFT 0.332 at all L.
- Dense plateau ~ 0.70 vs MFT ~ 0.80 — persistent, not a transient.

SSB: a robust order parameter



Why naive fails

Time-averaged $|\rho_1 - \rho_2|$ washed out: flips $(+,-) \leftrightarrow (-,+)$.

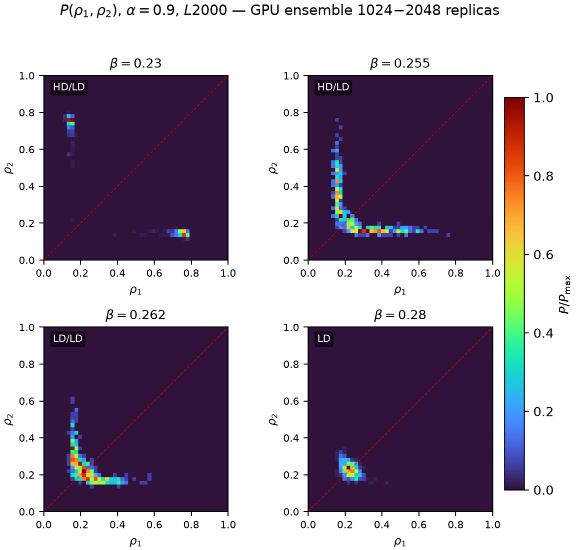
Robust: $\mathrm{std}(\rho_1 - \rho_2)$ over ensemble (512 reps).

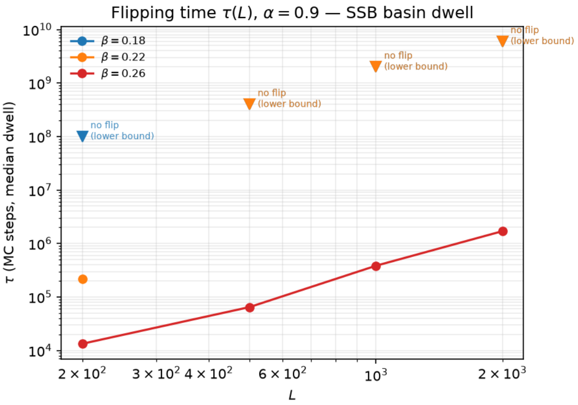
- $L = 1000$, $\alpha = 0.9$: $\langle \rho_{\mathrm{dense}} - \rho_{\mathrm{dilute}} \rangle$ $0.88 \to 0.36$.
- $\mathrm{std} \approx 0.12$–$0.17$ stays finite.
- SSB at low β (SIM ≈ 0.23).

$P(\rho_1, \rho_2)$ from an ensemble (paper Fig 3)

- $\alpha = 0.9$, $L = 2000$, 2048 replicas (2D heatmap, 48 bins): two peaks $\to$ arc $\to$ single — SSB visible at a glance.
- Full 3D in backup; animations:

3D mp4 2D mp4





Key message

SSB = metastability: basins live $\tau \sim e^{L/\xi}$ — at large L a run sees *one* basin.

- Zhu 2012 measured $\tau(L)$ for the *leaky* model ($p > 0$); we extend to $p = 0$ at α=0.9.
- Band edge (β=0.26): $\tau \approx 1.4{\cdot}10^4 e^{L/390}$ — $\times$100 per 1800 sites.
- Deep band: no flip in $6{\cdot}10^9$ steps at L=2000 ($\tau \gtrsim 10^{10}$).
- Explains why bands shrink: finite runs sample one basin $\Rightarrow$ equilibration needs $c \propto L$.

Simulations I: why BKL, why Fenwick

BKL rate-proportional (rejection-free)

- Pick next event $\propto$ rate — no rejections, exact Gillespie.
- Classic: scan active sites $\propto n_{\text{active}}$ — cheap at low β, costly at high density.
- Fenwick tree: binary indexed tree, $O(\log L)$ pick/update — $\sim$2.4$\times$ **faster, identical physics**.

Cost crosses in β: BKL wins dilute, Fenwick wins dense.

Kernel	Msteps/s (1 core)
classic BKL	1.8
Fenwick BKL	4.4

Why costly: equilibration $\propto cL^2$ steps — 10^8 at $L = 2000,\ c = 50$.

Simulations II: the GPU ensemble

One thread = one replica

$$\text{GPU}: \ \mathrm{rep}_1 \oplus \cdots \oplus \mathrm{rep}_N$$

- Serial chain → GPU is latency-bound ($\sim$ 35k/s/thread).
- Ensemble → GPU is perfect: 2048 replicas in parallel.
- Exactly what SSB needs: $P(\rho_1, \rho_2)$ is an ensemble distribution, not a time average.

Backend	Msteps/s agg.
CPU 1 thread (Fenwick)	4.4
CPU 30 workers	$\sim$100
GPU ensemble (RTX 4070)	120–180

Per-L bundles `results/<L>/` + `params.json`; plots read `.npz` —

no SIM at plot time, reproducible, crash-safe.

Outlook: what's next — beyond bulk MF

Coupling is the engine

Zhu *et al.* PRE **85**, 041132 (2012)

- Leaky entrance: inject at $p\alpha$ also if other lane's exit is occupied.
- p=0 our model, p=1 decoupled; SSB grows with coupling.
- Both broken phases die at $p_c \approx 0.6$ before p=1.
- Flipping time $\tau \propto e^L$: SSB genuine.

Is SSB real at p=0?

Tian *et al.* CPB **26**, 020503 (2017)

- N-cluster MF + current minimization: LD/LD should not exist (J highest in band).
- Boundary vs N decays $\sim e^{aN}$: $\beta_c(\infty) = 0.289$ at $\alpha = 0.9$ (MF 0.332, SIM $\approx$0.25).
- Yet their L=10^4 MC still sees LD/LD — verdict open.

Our next

- 2-site $\to$ 4-site cluster (Tian: N$\geq$5 adds only spurious sol. II).
- SIM to L = 4000–20000 at $c \approx 100$ (cL^2); track $\mathrm{std}(\rho_1 - \rho_2)$, $\tau(L)$.
- RNG at 10^{11} draws: counter-based streams (Philox; AES-CTR Trivium bank).

Take-away

Entrance coupling drives SSB; scaling + current minimization decide what survives.

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K. Zhu, N. Wang, Q.-Y. Hao, Y.-Q. Liu, R. Jiang, *Weakening interaction suppresses spontaneous symmetry breaking in two-channel asymmetric exclusion processes*, Phys. Rev. E **85**, 041132 (2012) — leaky entrance: SSB dies at $p_c \approx 0.6$; $\tau \propto e^L$.

S. Xiao, J. J. Cai, F. Liu, M. Z. Liu, *Effect of unequal injection rates and different hopping rates on asymmetric exclusion processes with junction*, Chin. Phys. B **19**, 090202 (2010) — Y-junction TASEP (different process).

B. Tian, R. Jiang, M. B. Hu, B. Jia, *Spurious symmetry-broken phase in a bidirectional two-lane ASEP with narrow entrances*, Chin. Phys. B **26**, 020503 (2017) — N-cluster + current minimization: LD/LD should not exist; MC at $L = 10^4$ sees it.

Open questions

- Do the broken bands collapse onto the first-order line as $L \to \infty$? Our L-scan is consistent with it; large L needs a budget $\propto L^2$.
- Does the MFT-vs-SIM boundary shift persist in the thermodynamic limit? It persists up to $L = 2000$ so far.
- Beyond MFT: two-site correlations / boundary-correlation parameters — can they close the gap?
- Extensions: asymmetric rates ($\alpha_1 \neq \alpha_2$), wider entrances, 3 channels.

Thank you!

Questions?

Backup slides

Anticipated questions — backups follow.

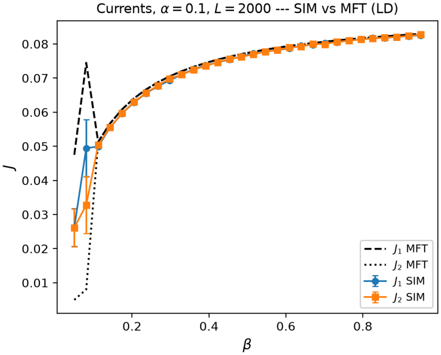
Main talk 60 min normal pace ($\sim$38 slides); backups not counted.

Backup: original paper vs our reproduction

Paper Fig. 2–3 are at $L \leq 1000$, $\sim 3\,\mathrm{k/site}$. Ours is $L = 2000$, $50\,\mathrm{k/site}$ — same topology, shifted boundaries, no speckle. We do not show scans in main to avoid clutter; this comparison is for fidelity questions.

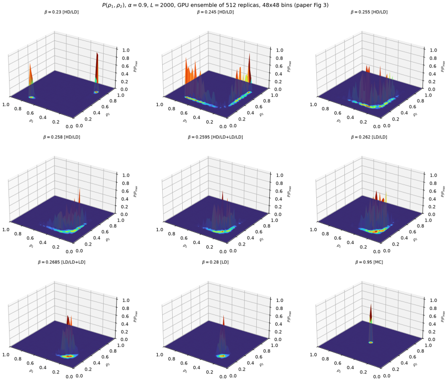
Pronina & Kolomeisky Fig. 2 (schematic) vs our Fig. 2 full (proj) — same phases, HD/LD band below MFT in both.

Backup: LD agreement — currents ($\alpha = 0.1$, $L = 2000$)



MFT (dashed) vs SIM (symbols) agree within s.e. across β — sanity check that the code and MFT are correct deep in LD. Deviation only near transitions.

Backup: 9-panel 3D $P(\rho_1, \rho_2)$ (original rendering)



3D view kept for completeness; main talk uses 2D heatmap (4 key β) for projector legibility. 2048 replicas, $\alpha = 0.9$, $L = 2000$.

Backup: equilibration must scale with L

- A fixed steps/site budget *under-equilibrates* large L: with 3×10^3/site, $L = 500$ gives dense $\sim$0.75 vs $L = 200$'s $\sim$0.92 — a systematic bias, not noise.
- Calibration probe: the SSB dense basin at $L = 1000$ needs $\sim 10^5$ steps/site to saturate (dense $\approx$0.89).
- Rule: steps/site $\approx c\,L$ with $c \approx 100 \Rightarrow$ total $\approx c\,L^2$.
- Today's runs use $c = 50$ (5×10^4/site); the fig-5 drift at $L \geq 4000$ is exactly this budget running out.

Backup: kernel performance details

- Classic BKL: cost $\propto n_{\text{active}}$ — cheap at low β (few events), expensive at high density.
- Fenwick: $O(\log L)$ always — wins at high β; the two cost curves cross in β.
- Per-thread GPU: $\sim$35k step/s (serial chain, latency-bound); $\sim$58–60k at $L = 8000$.

Backend	Mstep/s	rel.
CPU 1$\times$ classic	1.8	1$\times$
CPU 1$\times$ Fenwick	4.4	2.4$\times$
CPU 30 workers	$\sim$100	55$\times$
GPU (RTX 4070)	120–180	$\sim$90$\times$

GPU best for: phase diagrams, $P(\rho_1, \rho_2)$ ensembles, big statistics. Long single trajectories: CPU ($\sim$100$\times$ faster per-thread).